Network Layer: Control Plane

General Resources:

- NetworkingControlPlaneDefinitions.pdf
- NetworkingControlPlaneEquations.pdf

Topics Covered:

- Introduction to the Network-layer control plane
- Routing Algorithms
- Intra-AS Routing in the Internet: OSPF
- Routing Among the ISPs: BGP
- The SDN Control Plane
- ICMP: The Internet Control Message Protocol
- Network Management and SNMP, NETCONF/YANG

Practice Questions: Multiple Choice

Question 1

Which of the following statements correctly identify the differences between routing and forwarding. Select one or more statements.

- ☐ Forwarding refers to determining the route taken by packets from source to destination, and is implemented in the control plane.
- ☑ Routing refers to determining the route taken by packets from source to destination, and is implemented in the control plane.
- ☐ Forwarding refers to determining the route taken by packets from source to destination, and is implemented in the data plane.
- ☑ Forwarding refers to moving packets from a router's input to appropriate router output, and is implemented in the data plane.
- ☐ Routing refers to moving packets from a router's input to appropriate router output, and is implemented in the control plane.

- ☐ Routing refers to moving packets from a router's input to appropriate router output, and is implemented in the data plane.
- ☐ Forwarding refers to moving packets from a router's input to appropriate router output, and is implemented in the control plane.
- ☐ Routing refers to determining the route taken by packets from source to destination, and is implemented in the data plane.

Question 2

Match the name of the approach towards implementing a control plane with a description of how this approach works.

Answer List:

- A (typically) remote controller gathers information from routers, and then computes and installs the forwarding tables in routers.
- The network operator installs forwarding tables using the Simple Network Management Protocols (SNMP).
- Individual routing algorithm components - with a component operating in each and every router - interact with each other in the control plane. The individual routing algorithm component executing in a given router computes the local fowarding table fir that router.

Question List:

- Per-router control plane: **Individual routing algorithm components - with a component operating in each and every router - interact with each other in the control plane. The individual routing algorithm component executing in a given router computes the local fowarding table fir that router.**
- Software-defined networking (SDN): **A (typically) remote controller gathers information from routers, and then computes and installs the forwarding tables in routers.**

Question 3

What is the definition of a "good" path for a routing protocol? Chose the best single answer.

- ☐ A low delay path.
- ☑ Routing algorithms typically work with abstract link weights that could represent any of, or combinations of, all of the other answers.
- ☐ A path that has little or no congestion.
- ☐ A path that has a minimum number of hops.
- ☐ A high bandwidth path.

Question 4

Consider Dijkstra's link-state routing algorithm that is computing a least-cost path from node a to other nodes b, c, d, e, f. Which of the following statements is true. (Refer to Section 5.2 in the text for notation.)

- ☑ The values computed in the vector D(v), the currently known least cost of a path from a to any node v, will never increase following an iteration.
- ☐ Following the initialization step, if nodes b and c are directly connected to a, then the least cost path to b and c will never change from this initial cost.
- ☑ Suppose nodes b, c, and d are in the set N'. These nodes will remain in N' for the rest of the algorithm, since the least-cost paths from a to b, c, and d are known.
- ☑ In the initialization step, the initial cost from a to each of these destinations is initialized to either the cost of a link directly connecting a to a direct neighbor, or infinity otherwise.
- ☐ The values computed in the vector D(v), the currently known least cost of a path from a to any node v, will always decrease following an iteration.

Question 5

Match the name of a general approach to routing with characteristics of that approach.

Answer List:

- All routers have complete topology, and link cost information.

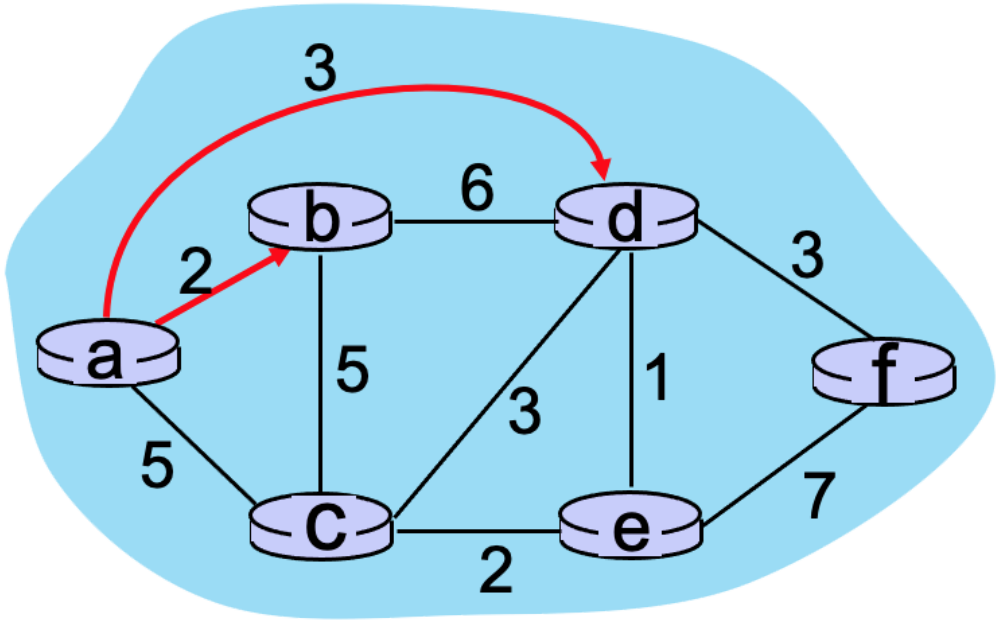
- Routing changes quickly over time.
- An iterative process of computation, exchange of informatoin with neighbors. Routers may initially only know link costs to directly-attached neighbors.
- Routes change slowly over time.

Question List:

- Centralized, global routing: All routers have complete topology, and link cost information.
- Decentralized routing: An iterative process of computation, exchange of informatoin with neighbors. Routers may initially only know link costs to directly-attached neighbors
- Static routing: Routes change slowly over time.
- Dynamic routing: Routing changes quickly over time.

Question 6

Consider the graph shown below and the use of Dijkstra's algorithm to compute a least cost path from a to all destinations. Suppose that nodes b and d have already been added to N'. What is the next node to be added to N' (refer to the text for an explanation of notation).

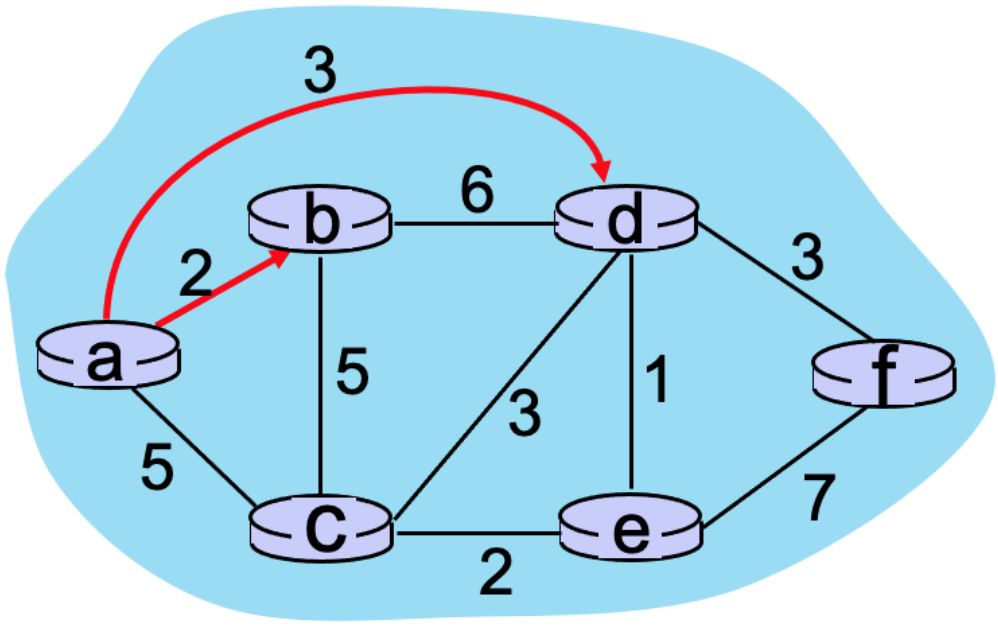


[Note: You can find more examples of problems similar to this here.]

- [x] e
- [ ] c
- [ ] f

Question 7

Consider the graph shown below and the use of Dijkstra's algorithm to compute a least cost path from a to all destinations. Suppose that nodes b and d have already been added to N'. What is the path cost to the next node to be added to N' (refer to the text for an explanation of notation).



- [x] 4
- [ ] 6
- [ ] 7
- [ ] 5

Question 8

Match the terms "interdomain routing" and intradomain routing" with their definitions. Recall that in Internet parlance, an “AS” refers to “Autonomous System” – a network under the control of a single organization.

Answer List:

- Routing among routers within same AS (“network”).
- Forwarding packets between two interfaces in different but adjacent subnetworks.
- Routing among different ASes (“networks”).

- Forwarding packets between two physically connected interfaces in a common subnetwork.

Question List:

- Interdomain routing: **Routing among different ASes (“networks”).**
- Intradomain routing: **Routing among routers within same AS (“network”).**

Question 9

Check the one or more of the following statements about the OSPF protocol that are true.

- ☐ The Open Shortest Path First (OSPF) Internet routing protocol implements a Bellman-Ford distance-vector routing algorithm.
- ☑ OSPF is an intra-domain routing protocol.
- ☐ OSPF is an interdomain routing protocol.
- ☑ OSFP uses a Dijkstra-like algorithm to implement least cost path routing.
- ☑ OSPF implements hierarchical routing

Question 10

Consider the OSPF routing protocol. Which of the following characteristics are associated with OSPF (as opposed to BGP)?

- ☐ Is an inter-domain routing protocol.
- ☑ Finds a least cost path from source to destination.
- ☐ Policy, rather than performance (e.g., least cost path), determines paths that used.
- ☑ Floods link state control information.
- ☑ Is an intra-domain routing protocol.

Question 11

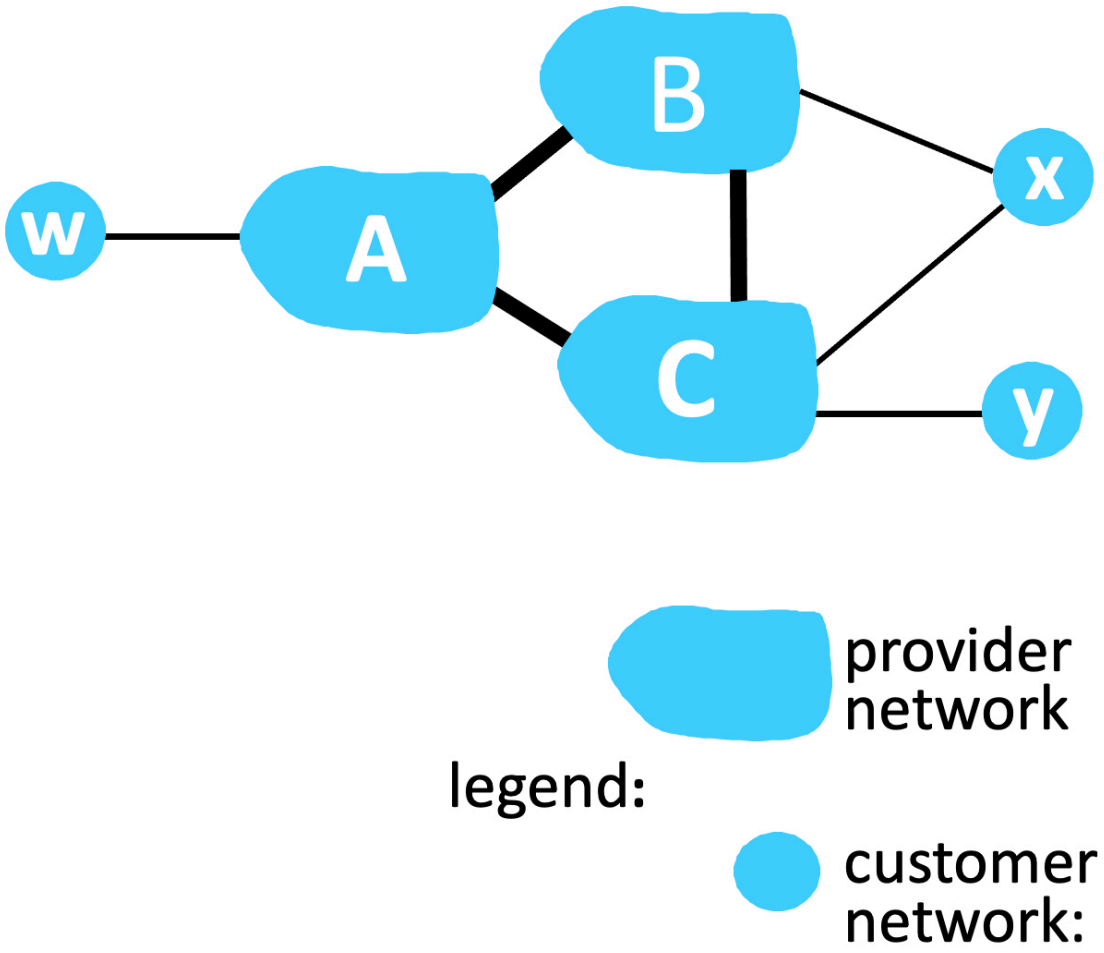
Among the following protocols, terminology or considerations, indicate those that are associated with "routing within a single network (typically owned and operated by one organization)."

- ☐ inter-AS routing

- ☐ BGP
- ☑ intra-domain routing
- ☐ inter-domain routing
- ☑ Driven more by performance than by routing policy
- ☑ intra-AS routing
- ☐ Driven more by routing policy than end-end routing performance
- ☑ OSPF

Question 12

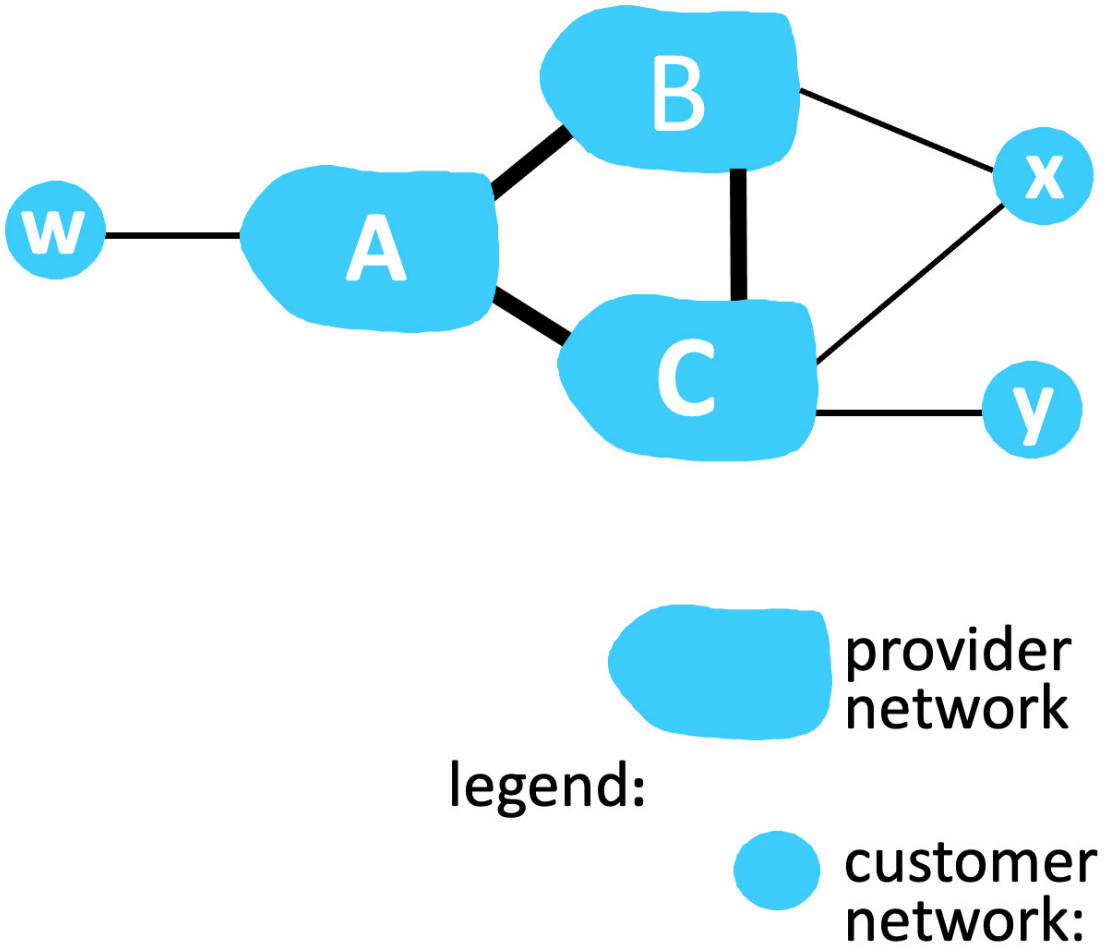
Suppose a provider network only wants to carry traffic to/from its customer networks (i.e., to provide no transit service), and customer networks only want to carry traffic to/from itself. Consider the figure below. To implement this policy, to which of the following networks would network C advertise the path Cy?



- [x] B
- [x] x
- [x] A
- [ ] w

Question 13

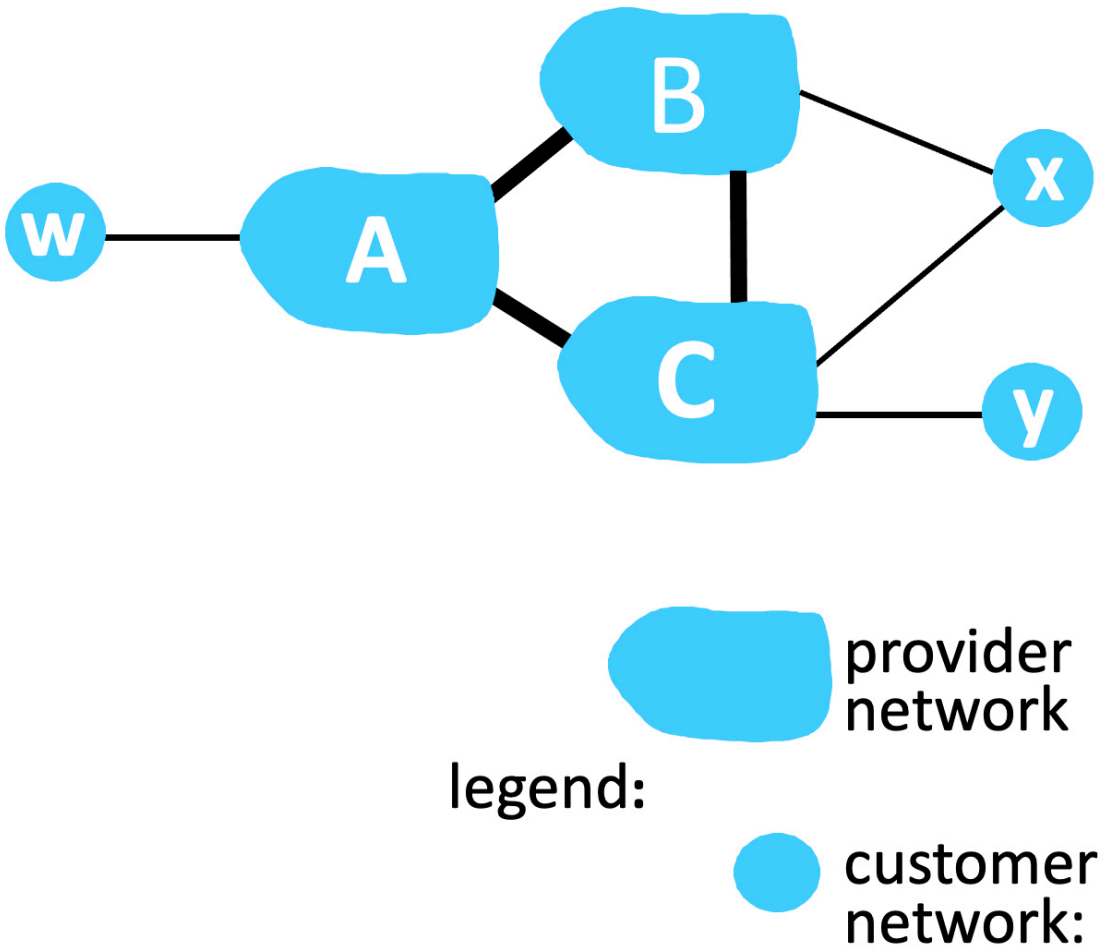
Again, suppose a provider network only wants to carry traffic to/from its customer networks (i.e., to provide no transit service), and customer networks only want to carry traffic to/from itself. Suppose C has advertised path Cy to A. To implement this policy, to which of the following networks would network A advertise the path ACy?



- ☐ B
- ☐ x
- ☑ w
- ☐ C

Question 14

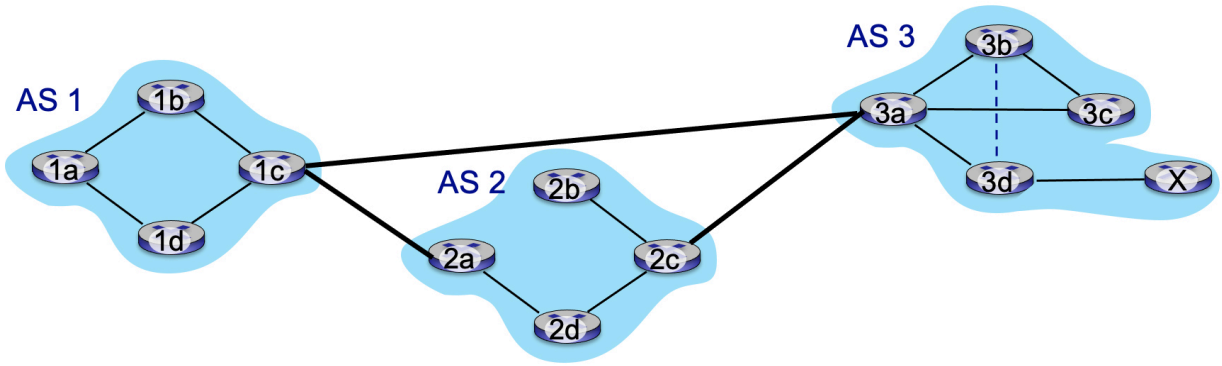
Again, suppose a provider network only wants to carry traffic to/from its customer networks (i.e., to provide no transit service), and customer networks only want to carry traffic to/from itself. Suppose C has advertised path Cy to x. To implement this policy, to which of the following networks would network x advertise the path xCy?



- ☐ A
- ☐ B
- ☐ w
- ☐ None of these other networks
- ☐ C

Question 15

Consider routers 2c and 2d in Autonomous System AS2 in the figure below. Indicate the flavor of BGP and the router from which each of 2c and 2d learns about the path to destination x.



Answer List:

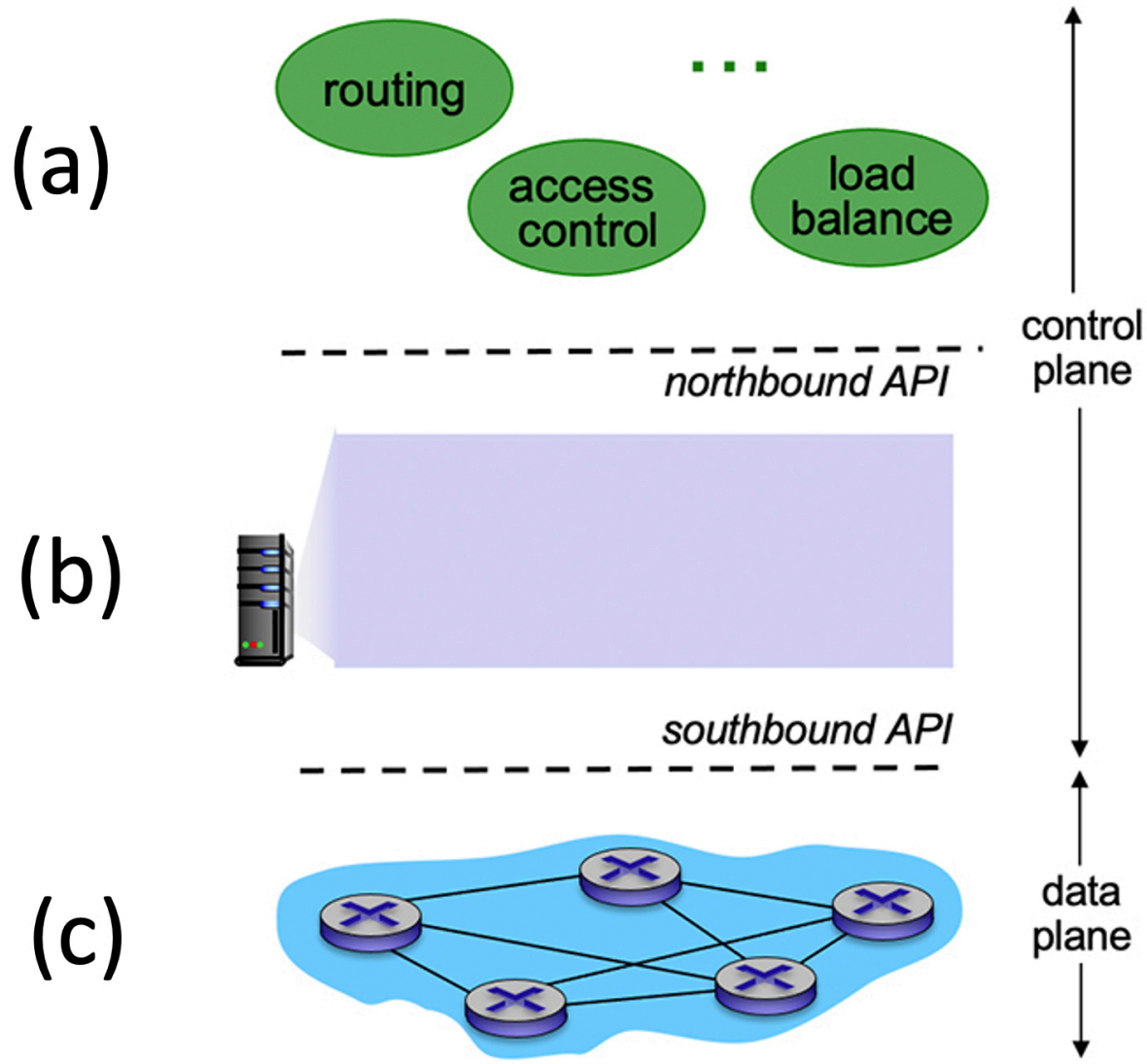
- From x via eBGP.
- From 3a via eBGP.
- From 2c via eBGP.
- From 2c via iBGP.
- From 3a via iBGP.

Question List:

- How does router 2c learn of the path AS3, X to destination network X: **From 3a via eBGP.**
- How does router 2d learn of the path AS3, X to destination network X: **From 3a via iBGP.**

Question 16

Consider the SDN layering shown below. Match each layer name below with a layer label (a), (b) or (c) as shown in the diagram.



Answer List:

- (a)
- (b)
- (c)

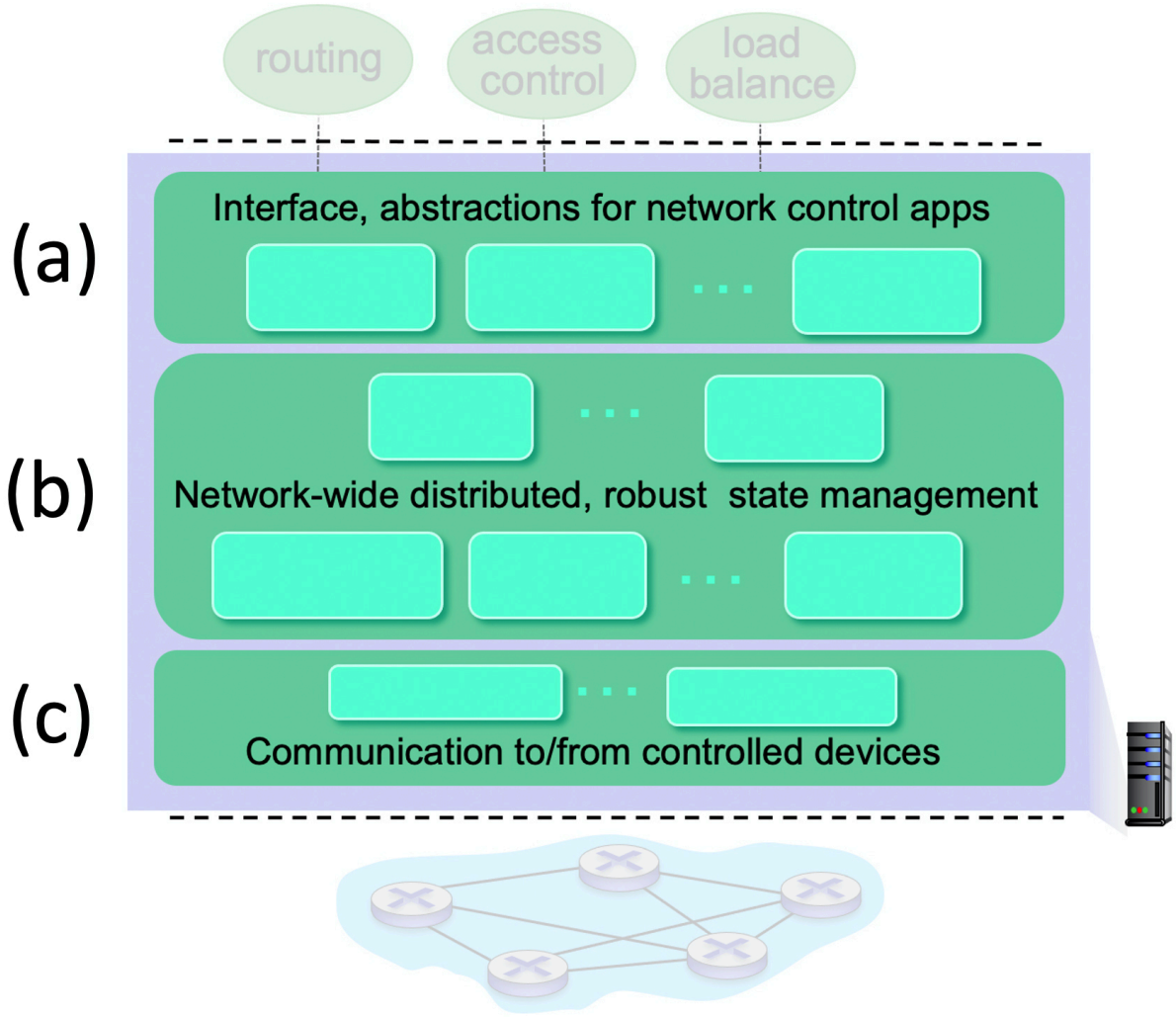
Question List:

- SDN Controller (network operating system): B
- SDN-controlled switches: C

- Network-control applications: A

Question 17

Which of the functions below belong in the controller layer labeled "Interface, abstractions for network control apps"? Check all below that apply.



- ☐ Intent
- ☐ Link-state information
- ☐ Switch information
- ☐ OpenFlow protocol

- ☐ Network graph
- ☐ Statistics
- ☐ Flow tables
- ☐ Host information

Question 18

Which of the statements below about ICMP are true?

- ☐ ICMP messages are carried in UDP segments using port number 86.
- ☑ ICMP messages are carried directly in IP datagrams rather than as payload in UDP or TCP segments.
- ☐ ICMP communicates information between hosts and routers by marking bits in the IP header.
- ☑ ICMP is used by hosts and routers to communicate network-level information.
- ☑ The TTL-expired message type in ICMP is used by the traceroute program.